Generative AI Meets Open-Ended Survey Responses: Research Participant Use of AI and Homogenization

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Abstract

The growing popularity of generative AI tools presents new challenges for data quality in online surveys and experiments. This study examines research participants' use of large language models to answer open-ended survey questions and describes empirical tendencies in human vs LLM-generated text responses. In an original survey of participants recruited from a popular online platform for sourcing social science research subjects, 34% reported using LLMs to help them answer open-ended survey questions. Simulations comparing human-written responses from three pre-ChatGPT studies with LLM-generated text reveal that LLM responses are more homogeneous and positive, particularly when they describe social groups in sensitive questions. These homogenization patterns may mask important underlying social variation in attitudes and beliefs among human subjects, raising concerns about data validity. Our findings shed light on the scope and potential consequences of participants' LLM use in online research.

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1 Introduction

The rise of web-based, self-administered surveys and online experiments in social science research has created valuable new opportunities for researchers but raises concerns about data quality. Issues such as participant inattention, disengagement, and dishonesty are often amplified in these formats (Alvarez et al., 2019; Ternovski and Orr, 2022; Bell and Gift, 2023). Popular online platforms like Amazon Mechanical Turk and Prolific present additional challenges, including automated bots, multiple submissions, and cross-contamination between studies (Chmielewski and Kucker, 2020; Douglas et al., 2023). Researchers have worked to measure these issues and develop mitigation strategies to improve data quality (Berinsky et al., 2014; Cuskley and Sulik, 2024).

In this article, we add to this work by examining an emerging challenge in online data collection: research participants' use of generative AI. The release of low-cost, public-facing AI tools like ChatGPT has spurred their rapid adoption in professional settings and among the general public. Powered by large language models (LLMs), these commercial AI tools have exhibited strong performance on tasks such as producing natural-sounding original text, offering writing assistance, and generating ideas. Scholars have already started examining the implications of generative AI tools in arenas like education and the workplace (e.g. Smolansky et al., 2023; Dell'Acqua et al., 2023; Brynjolfsson et al., 2023).

This article extends these lines of inquiry to generative AI's implications for web-based research. We examine participants' use of AI when answering open-ended text questions, a place where text-generation tools may prove especially disruptive. Open-ended questions appear widely across social science study designs. In surveys and experiments – the focus of the present paper – they are used to deliver interventions or experimental treatments, measure outcomes, probe the reasons or mechanisms behind close-ended choices, and assess the validity of other measures (Ziegler, 2022; Ferrario and Stantcheva, 2022; Hadler, 2023; Landesvatter and Bauer, 2024). Open-ended responses are also a popular object of analysis in the literature on text as data (Roberts et al., 2014; Egami et al., 2022) and used in computational social science for training and evaluating machine

learning models. While valuable for researchers, study participants report finding open-ended questions burdensome and time-consuming (Fowler et al., 2023). Online study participants, who usually receive flat payments as compensation, are incentivized to finish studies quickly to maximize their earnings. Against this backdrop, emerging evidence from crowdsourcing platforms suggests that some online research participants are turning to generative AI to answer writing-intensive questions (Veselovsky et al., 2023a,b).

This article offers a first descriptive portrait of participants' use of LLMs to answer open-ended questions. We fielded a survey on Prolific, a popular online survey platform, to measure the self-reported prevalence of LLM use, understand participants' reasons for using or not using LLMs, characterize the prompting strategies participants employ, and describe variations in LLM use among subgroups.

We then investigate the potential consequences of this emerging practice for the quality, content, and variance of survey responses, with the aim of informing researchers' expectations about how participants' LLM use may shape study results. Extant research suggests participants' use of LLMs may raise several potential issues. Most immediately, when participants generate responses to open-text questions, those answers can diverge from what respondents would have written absent AI tools. While some scholarship on how well carefully fine-tuned LLMs can predict survey answers has suggested LLM-generated responses are sometimes consistent with average human responses (Kim and Lee, 2023; Bisbee et al., 2024; Kozlowski et al., 2024; Ashokkumar et al., 2024), it is far less clear whether the ways in which research participants use off-the-shelf generative AI tools yield responses that align with their actual perspectives, rather than those encoded as generic positions in LLMs. Other research suggests that LLMs – whether as text generators or as writing assistants – tend to produce or steer people towards answers that are more homogeneous in style and content than natural human writing (Padmakumar and He, 2023; Anderson et al., 2024). Consequently, LLM-influenced study responses may mask important forms of social variation.

We evaluate these possibilities by assessing the extent to which LLM-produced responses align with and diverge from human-generated responses. To do so, we draw on open-ended questions

posed in three studies fielded as part of the Time-Sharing Experiments of the Social Sciences (TESS) program prior to the public release of ChatGPT. We compare those responses to LLM-simulated responses to the same questions. Our simulations are informed by our survey, in which we asked online survey takers how they would prompt an AI chatbot to answer each of the three questions.

We find that 34% of Prolific participants report using LLMs at least some of the time when they answer open-ended questions, with 9% reporting using LLMs "most of the time" or "always." We also find that LLM usage varies along demographic dimensions that correlate with cleavages in public opinion, which could bias estimates of between-group differences. Specifically, Black, male, college-educated, Republican, and participants newer to the Prolific platform are more likely to report using LLMs in our survey, though these patterns may change as AI technology diffuses more widely. Those who use LLMs most commonly cite needing help expressing their thoughts, AI's speed and writing quality, and researchers providing unclear instructions as reasons for turning to LLMs. Their most common prompting strategies include asking AI to generate ideas or provide additional information on the topic. Notably, perspective-taking prompts that incorporate participants' pre-existing views or social positions are rare. Meanwhile, those who refrain from using LLMs often cite moral reasons related to authenticity, ethical concerns about "cheating," or compliance with rules.

Comparing LLM-generated responses with human responses recorded in TESS data, we find that LLM-written text has a larger vocabulary but exhibits greater homogeneity across responses. The substance and tone of LLM-generated text also depart sharply from human-written answers, with LLM-generated answers using more positive language. In sensitive questions about group perceptions and stereotypes, human-written responses are more likely to feature dehumanizing language than LLM-simulated responses. Overall, our results show that free-text survey responses written by humans convey a broader range of views than answers generated by LLMs.

Together, our results provide an initial view of the emerging landscape and the consequences of study participants' use of LLMs. We present data on the scope and distribution of this practice

among study participants. Drawing on simulated LLM answers, we outline ways AI use could affect conclusions drawn from online surveys and experiments, while identifying directions for future research to further investigate the consequences of this evolving development. We conclude by identifying implications for research practice and connecting the study's results to related literatures on silicon sampling, AI evaluation, and the societal impacts of AI.

2 Generative AI and Human Subjects Surveys and Experiments

As generative AI use spreads among the general public, scholars and observers have raised questions about the implications of these tools for how people write and express themselves across many domains of social life, from peer review to medicine to the arts (Epstein et al., 2023; Liang et al., 2024; Tai-Seale et al., 2024). Recent studies of crowd-working platforms that social scientists often rely on to recruit research participants suggest that the consequences of LLMs extend to how participants engage with online studies. For instance, two studies of participants on Amazon Mechanical Turk (MTurk) and Prolific found that substantial shares – between 33-46% depending on how one infers LLM usage – relied on an LLM to complete a text summarization task (Veselovsky et al., 2023a,b). In a survey of crowdworkers fielded in December 2023, 13% of Prolific, 15% of Clickworker, 73% of MTurk participants reported ever using an AI chatbot to complete a crowdsourcing task (Christoforou et al., 2024). This article builds on this prior work measuring AI use in crowd-working tasks broadly, focusing specifically on the implications of AI use in social science surveys and experiments where it can influence the measurement of human subjects' subjective beliefs, attitudes, experiences, and behaviors.

Research participants' use of AI shares similarities with past challenges researchers have faced in online studies, but also presents unique considerations. LLM usage resembles previously examined issues such as bots in online surveys (Chmielewski and Kucker, 2020) and respondents' information search behaviors during studies (Clifford and Jerit, 2016; Gummer and Kunz, 2022; Graham, 2023). Like these earlier challenges, LLM use consists of an external influence on study

participation that can undermine the authenticity of participants' responses and compromise the validity of research findings. Researchers have developed various solutions for these past issues, such as attention checks and response time analysis (Berinsky et al., 2014; Kennedy et al., 2020), which may inform strategies for addressing LLM use. However, LLMs are also distinct in important ways. Unlike bots, which are typically deployed to game survey systems, LLMs can be used by real human participants as assistive tools and raise novel questions about the authenticity and validity of responses. Further, while information search traditionally involves participants actively looking up external content to inform their answers, LLMs can generate answers with less active effort on the part of participants.

In this article, we investigate the consequences of subjects' LLM use by focusing on the case of open-ended text questions, writing-intensive components of studies for which participants might be most likely to turn to LLMs. When open-ended text responses are used for measurement – whether to capture outcomes or mechanisms, or to probe other answers further – participants' use of LLMs raises important questions about how they might shape the natural responses that participants would give (Davidson, 2024).

In the ideal, LLMs could offer benefits that are aligned with researchers' objectives of obtaining valid, unbiased results. AI writing assistance could make it easier and faster to write out fleshed-out responses or help participants more precisely express their true underlying beliefs or attitudes. Research on the implications of LLM use in other domains, for instance, suggests that LLMs may be especially beneficial for study participants with weaker writing skills (Noy and Zhang, 2023; Mollick, 2023; Brynjolfsson et al., 2023). To the extent that LLMs lower burdens on participant time and effort without compromising measurement validity, LLM use among participants may generate other ancillary benefits, like minimizing respondent fatigue, item non-response, and survey attrition (Peytchev, 2009; Zuell et al., 2015).

But emerging evidence on the values and perspectives embedded in LLMs raises the possibility that participants' interactions with them could distort answers and compromise data quality. If participants directly copy LLM-generated responses or craft answers informed by LLM suggestions

(Jakesch et al., 2023), how might that influence answers gathered in response to open-ended survey questions? A rapidly growing body of research has aimed to characterize LLM tendencies when prompted to respond to attitudinal and behavioral questions ordinarily posed to humans. While much of this work was developed to address other concerns, such as probing the social biases or point of view that LLMs encode (Santurkar et al., 2023; Motoki et al., 2024) or evaluating how well LLMs can simulate human attitudes and actions in "silicon samples" (Boelaert et al., 2024; Bail, 2024), it offers important clues as to how study participants' exposure to LLM assistance could impact their answers in surveys and experiments.

A key lesson from this prior work is that LLM-provided answers to survey questions can be highly sensitive to how LLMs are prompted (Wright et al., 2024). Users may prompt LLMs, for instance, to offer a generic position on a question, such as "What is a reason to support or oppose gun control?" Studies on the default positions that publicly available, commercial LLMs tend to generate find that they most closely reflect the positions of particular subpopulations, such as people in the US and some countries in Europe and South America (Durmus et al., 2023) and more left-leaning individuals (Santurkar et al., 2023; Hartmann et al., 2023; Martin, 2023; Rozado, 2023; Motoki et al., 2024). This suggests that, when prompted in this manner, LLM-assisted answers may steer some respondents away from their true positions and toward the generic views of LLMs.

Users may also prompt LLMs in more tailored ways, such as by signaling their own stances on an issue or providing additional detail on their backgrounds, such as "I'm a progressive woman from the Midwest. What is a reason to support gun control?" This latter approach approximates a common strategy that researchers employ to study how accurately LLMs can predict human responses. In such studies, researchers typically supply LLMs with background information about a hypothetical respondent to assemble a persona that is then used to simulate responses. Some studies suggest that LLMs can reasonably match average human opinions and positions in specific contexts (Argyle et al., 2023; Kozlowski et al., 2024; Kim and Lee, 2023; Bisbee et al., 2024; Ashokkumar et al., 2024). Closest to our setting, Argyle et al. (2023) show that when prompted

with sociodemographic profiles of hypothetical respondents, LLMs produce similar words to describe Republicans and Democrats as do real respondents matching those demographics. Other studies meanwhile suggest that LLMs can exhibit tendencies toward misportraying or caricaturing the positions of members of distinct social groups (Wang et al., 2024; Cheng et al., 2023; Durmus et al., 2023; Neumann et al., 2024) [1] Overall, however, this literature suggests that such persona-based prompting may lead to AI-informed answers that are closer to participants' true answers than would more generic prompting strategies.

Yet, despite the significance of prompting strategies, we know little about how study participants engage with AI when answering open-ended questions. In this study, we address this gap by directly asking study participants who use AI to draft sample prompts they would use to help them answer survey questions. While researchers use sophisticated methods to simulate answers, including careful prompt engineering and fine-tuning to mimic human responses (Ollion et al., 2023; Jensen et al., 2022), most participants are unlikely to engage with AI in such tailored or detailed ways. For many, drafting such prompts may be more burdensome than simply answering the question directly. By examining the types of responses that participant-written prompts produce, this study reveals what participants are exposed to when relying on AI, shedding light on the direction in which AI interactions may influence participant answers.

Beyond congruence between the average or typical responses of LLMs and humans, additional research highlights the possibility that LLMs may suppress natural human variation, homogenizing or standardizing respondents' answers in both content and style. LLMs embed the perspectives present in their training data, as well as the commitments and concerns of their developers (Bender et al., 2021; Alvero and Peña, 2023). Despite efforts by LLM developers to "democratize" design decisions and incorporate diverse text in their training corpora, LLMs commonly learn and reflect dominant cultural styles and perspectives, such as those held by demographic majorities or groups over-represented in cultural production (Cao et al., 2023; Naous et al., 2024; Tao et al., 2024; Gillespie, 2024). It is also the case that many LLMs, including the most popular platforms where

[1] We focus here on methodological challenges to using LLMs to simulate opinions, but see Agnew et al. (2024) and Wang et al. (2024) for ethical arguments.

users can interface with them via chatbot, are not scientific but rather commercial products: having a product that can give consistent, reliable results may be better for some business and industrial scientific applications. However, it can present a hazard for surveys and experiments where LLM-induced homogenization can lead to misleading or incorrect conclusions.

A variety of studies have highlighted the potential of LLMs to homogenize people's positions and written self-expressions. LLMs tend to predict that opinions and attitudes within social groups are more uniform than they are in reality (Santurkar et al., 2023; Park et al., 2024; Wang et al., 2024; Bisbee et al., 2024). They also tend to produce writing with particular styles and word choices. Wright et al. (2024), for example, show that LLM open responses to political questions often repeat similar linguistic tropes, even across prompting strategies, models, and political stances. Another study comparing human-written and LLM-generated college admissions essays also finds that the writing styles of LLMs exhibit much less variation than the writing styles of humans within different social groups (Alvero et al., 2024). Liang et al. (2024) similarly show that academic peer reviews with linguistic markers consistent with being AI-generated tend to be more homogeneous and formulaic. Experimental research on the effects of LLM-based writing assistants further offers evidence that LLMs steer human authors toward writing opinion essays and stories that are less diverse (Padmakumar and He, 2023; Doshi and Hauser, 2024). Thus, we might expect that LLM use among research participants – whether to generate answers directly or to inform human-written answers – could lead to data that obscures social variation.

In sum, while the growing body of research on LLMs provides important insights into the capabilities and limitations of these tools, there remains a significant gap in understanding how their use by research participants might influence the quality and validity of textual data collected in online surveys and experiments. Our study addresses this gap by providing the first empirical portrait of LLM use among online study participants, examining its prevalence, participants' motivations for using LLMs, and their prompting strategies when seeking AI assistance. Furthermore, we extend prior research by simulating LLM-generated responses based on the prompting strategies reported by actual online research participants, rather than using researcher-designed prompts. By com-

paring these simulated responses to human answers from pre-ChatGPT surveys, we offer novel insights into how participants' LLM use may impact the content, style, and variance of responses to online studies.

3 Overview of Analytic Approach and Data

Our analysis proceeds in two parts. Part I provides a descriptive portrait of AI use among online study participants, drawing on results from an original survey of individuals active on Prolific, a popular online platform for recruiting study subjects. Part II compares the characteristics of human vs. LLM-generated open-ended survey responses. Specifically, we examine responses from three TESS survey questions fielded before the public release of ChatGPT against LLM-generated answers produced using prompts crafted by participants in our Prolific survey. In what follows, we describe our primary data sources. We further detail our methods and measures in Analysis Parts I and II.

3.1 Survey on AI Use

We conducted an original survey to investigate how online survey participants engage with LLMs when answering open-ended questions. Since the target population for this survey is individuals who have experience taking online surveys and are likely to do so in the future, we recruited directly from Prolific, a popular digital platform for sourcing survey and experiment participants. As online non-probability-based samples continue to be used widely across the social sciences due to their cost-effectiveness and ease of access (Coppock and McClellan, 2019; Weinberg et al., 2014), the behavior of online participants – who may also have the strongest motivation for using AI to help them work more efficiently – is especially important to understand.

To maximize coverage and variation across key sociodemographic characteristics, we used Prolific's "representative sample" recruitment option, which sets quotas based on ethnicity, gender, age, and political affiliation to achieve a sample composition that mirrors that of the U.S.

population on these dimensions. While quota sampling approaches remain fundamentally different from probability-based sampling, they are often the best available strategy for online studies and are thus widely used (Yang and Banamah, 2014). We recruited 799 participants to our study; our final analytic sample excludes two participants who did not give consent, two participants who did not pass an attention check embedded halfway through the survey, and one participant who was flagged as a bot by a CAPTCHA test.

Table 1 shows the characteristics of the sample. Compared to the U.S. population, the sample has a smaller share of Hispanic/Latino individuals, is more college-educated, and is slightly older. Almost two-thirds of the participants joined Prolific within the past year, and the vast majority earn most of their income from sources other than the platform. However, more than 80 percent of participants reported taking at least three other surveys in the 24-hour period before taking our survey, suggesting that most are regular survey-takers.

Table 1: Prolific Sample Descriptive Statistics (N = 794)

	% or Mean (Std. Dev)
Race (categories not mutually exclusive)	
White	70.5
Hispanic/Latino	11.6
Black/African-American	16.4
Asian/Asian-American	8.4
Other	5.8
Gender	
Female	50.0
Male	47.7
Non-binary or other	2.3
Age	45.2 (16.3)
Educational Attainment	
Less than BA	44.2
BA or higher	55.8
Political Party Affiliation	
Democrat	33.1
Independent	37.3
Republican	28.6
Other	1.0
Time on Prolific	
Less than 6 months	54.7
6 months - a year	9.2
More than a year	36.1
Share of income from Prolific	
Less than half	83.4
More than half	16.6
Num. surveys taken in previous 24 hrs	
Fewer than 3	17.0
3-6	42.8
7 or more	40.2

Source: Subjects recruited on Prolific in summer 2024

Our main outcome of interest was the frequency with which participants used LLMs ("AI assistants or chatbots") for help with answering open-ended survey questions. To minimize social desirability bias, we framed the survey as a study about online workers' use of and expertise with "new AI technologies." Given that some survey respondents might still feel pressure to misrepresent their LLM usage out of reputational concerns, we emphasized that their answers wouldn't

influence their compensation or eligibility for future studies. We also assuaged these concerns by not collecting Prolific IDs on the survey and reassuring respondents that their personal identifiers would not be shared with others.

Additionally, we asked respondents to identify reasons for using (or not using) AI and probed *how* they would prompt or talk to AI assistants or chatbots to receive help with writing. To do this, we showed participants the exact questions seen by TESS participants in the three focal studies. These respondent-generated prompts provide insight into how survey participants actually engage with LLMs to generate free-text responses, and we use them to simulate the LLM responses in our study.

3.2 Data on Human-Generated Survey Responses

For a source of human-crafted answers to open-ended survey questions, we use survey experiment data from three studies conducted via the Time-sharing Experiments for the Social Sciences (TESS) program, which allows researchers to field studies on NORC's nationally representative, probability-based AmeriSpeak panel. All three studies were fielded prior to the public release of ChatGPT on November 30, 2022. The data for these studies is publicly available through TESS's Open Science Framework repository. Table 2 summarizes the questions we used and their sample sizes. We chose these three surveys because they were among the last waves of TESS studies administered before ChatGPT's public release, featured open-ended questions that were reasonably easy to isolate and replicate, and addressed topics of theoretical interest to the measurement of polarization and homogenization in public opinion data. Two of the studies – Jardina et al. (2021) (JSD for short) and Graham (2021) – explicitly ask about attitudes toward well-defined social groups and are especially germane to debates about measuring group stereotypes and sensitive topics. The third study, Groenendyk et al. (2025), focuses on a more abstract topic, which may be more deliberative and cognitively demanding for study participants. Collectively, the three questions in these studies capture some of the common purposes for open-ended survey questions in the social sciences.

We simplified some study designs to facilitate cleaner comparisons between simulated LLM responses and human-written responses. For Graham, we examine responses from those randomly assigned to write about political parties (as opposed to their favorite activity). For Groenendyk, we limited the sample to those who were not exposed to randomly assigned definitions of "politics" earlier in the study.

Table 2: **TESS Questions.** Sample size is for the open-ended question examined and excludes missing/invalid values in the TESS replication data.

Study & Principal Investigator(s)	**Question**	**Sample Size**	**Field Period**
"The Influence of the Terms We Use on Measures of Racial Attitudes" Ashley Jardina and LaFleur Stephens-Dougan	In the text box below, briefly describe what comes to mind when you think of [randomized: blacks | African Americans] in American society today.	1,928	05/25/2020-03/23/2021
"Conditional Commitment? Partisan Identity and Support for Democracy in the United States" Matthew Graham	What makes you similar to [own party: Democrats | Republicans]? What makes you different from [other party: Democrats | Republicans]?	1,930	08/31/2019-01/21/2020
"What Motivates Interest in Politics?" Eric Groenendyk	What is it about politics that makes you [uninterested | interested]?	346	04/13/2021-07/01/2021

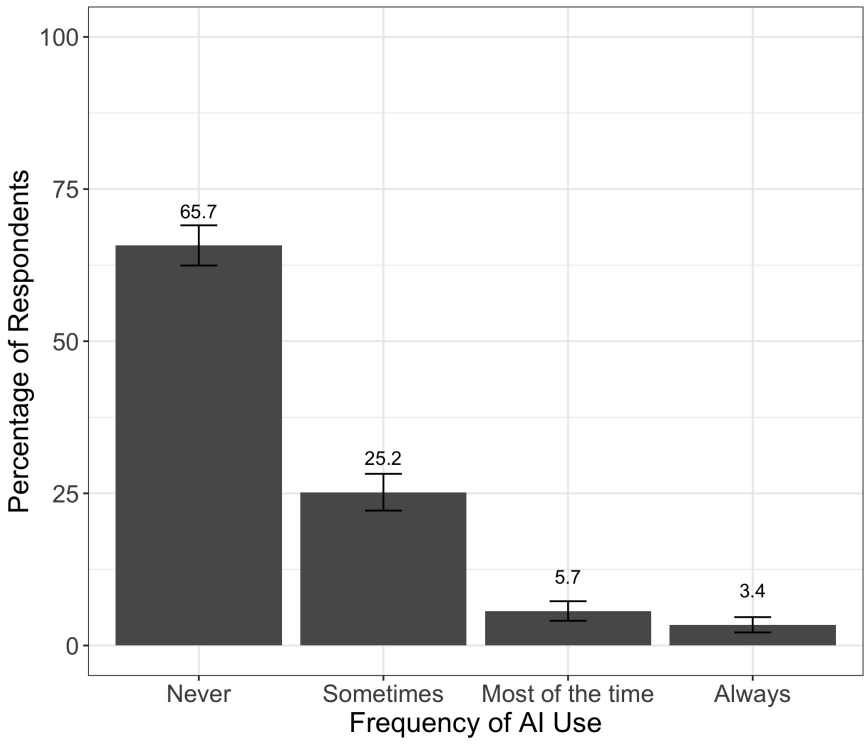
4 Analysis Part I: Portrait of LLM Use Among Research Participants

In Analysis Part I, we summarize findings from our survey of Prolific study participants on their use of AI when participating in studies.

4.1 Patterns in AI Use

Figure 1 shows the frequency of LLM usage among survey respondents. About two-thirds of respondents state that they have never used LLMs to help them answer open-ended survey questions. About one-quarter of participants report that they "sometimes" use AI assistants or chatbots for help with writing. Less than 10% of the sample reports using LLMs very frequently ("most of the time" or "always") when answering questions that require writing. Overall, the results suggest that a share of online survey participants are now relying on generative AI for help with writing responses to open-ended questions, but LLMs have not reached widespread adoption.

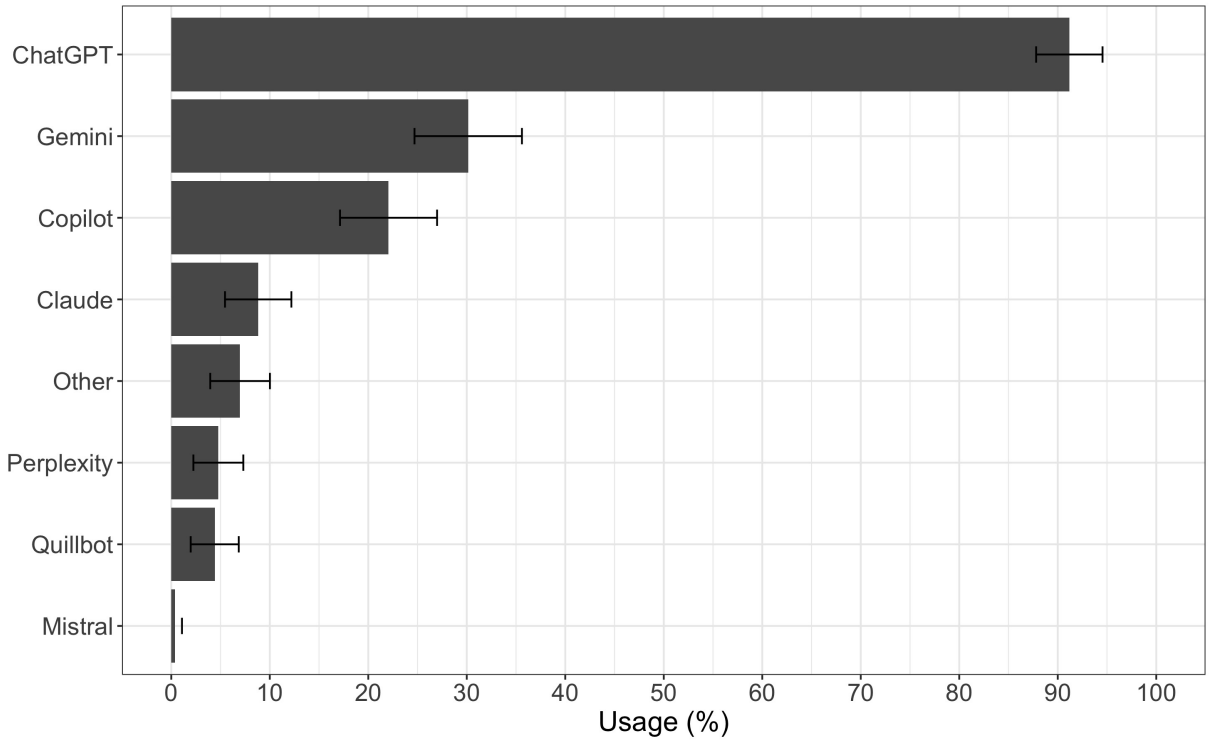
Figure 1: **LLM Usage** among all respondents



As shown in Figure 2, OpenAI's ChatGPT is by far the most popular model among respondents who report using LLMs for writing assistance at least some of the time. Over 90% of AI users rely on ChatGPT for writing assistance. About 30% of AI users in the sample work with Google Gemini and about 22% use Microsoft Copilot, but the vast majority (>80%) of Gemini and Copilot users also report using ChatGPT. Most AI users only rely on one model, though a small minority (about

4%) reported 4-5 different LLMs when asked to select all the AI tools they use.

Figure 2: **Most common LLMs** used by respondents who reported using AI "sometimes," "most of the time," or "always" to complete open-ended writing questions. Respondents were asked to select all options that applied.



Are certain subpopulations of survey participants more likely to use LLMs than others? Subgroup differences in LLM usage have important consequences for both within-group inferences and between-group inferences. For example, if members of a social group are more likely to use LLMs than the rest of the population, their survey responses may be more similar to each other, creating the false impression that individuals of this group are a homogenous bloc. Moreover, members of groups with higher LLM usage may hold views that diverge from the rest of the population's in reality, but produce LLM-generated survey responses that misleadingly bear greater resemblance to the population average. This effectively suppresses the group's opinion, which would bias population-level mean and variance estimates. Table 3 shows how the subset of LLM users differs from those who say that they never use AI assistants or chatbots for help with answering open-ended questions. In this first snapshot of LLM usage, we find that Black, male,

college-educated, and Republican respondents are more likely to use LLMs for help with open-ended questions. Those newer to the platform – people who started working on Prolific in the past 6 months – are also more likely to use LLMs for writing assistance than participants who have been taking surveys for at least a year.

Table 3: Sample Characteristics by AI Usage

	Uses AI (*N* = 272)	**Never uses AI (*N* = 522)**
	% or Mean (Std. Dev)	% or Mean (Std. Dev)
Race (categories not mutually exclusive)		
White	59.9	76.1
Hispanic/Latino	11.4	11.7
Black/African-American	26.8	10.9
Asian/Asian-American	6.6	9.4
Other	5.9	5.7
Gender		
Female	43.8	53.3
Male	55.5	43.7
Non-binary or other	<1.0	3.1
Age	44.3 (16.0)	45.8 (16.4)
Educational Attainment		
Less than BA	36.0	48.5
BA or higher	64.0	51.5
Political Party Affiliation		
Democrat	28.3	35.6
Independent	32.4	39.8
Republican	37.5	23.9
Other	1.8	0.6
Time on Prolific		
Less than 6 months	64.0	49.8
6 months - a year	8.8	9.4
More than a year	27.2	40.8
Share of income from Prolific		
Less than half	84.2	83.0
More than half	15.8	17.0
Num. surveys taken in previous 24 hrs		
Fewer than 3	17.3	16.9
3-6	38.2	45.2
7 or more	44.5	37.9

Source:Subjects recruited on Prolific during summer 2024

4.2 Reasons For and Against Using AI

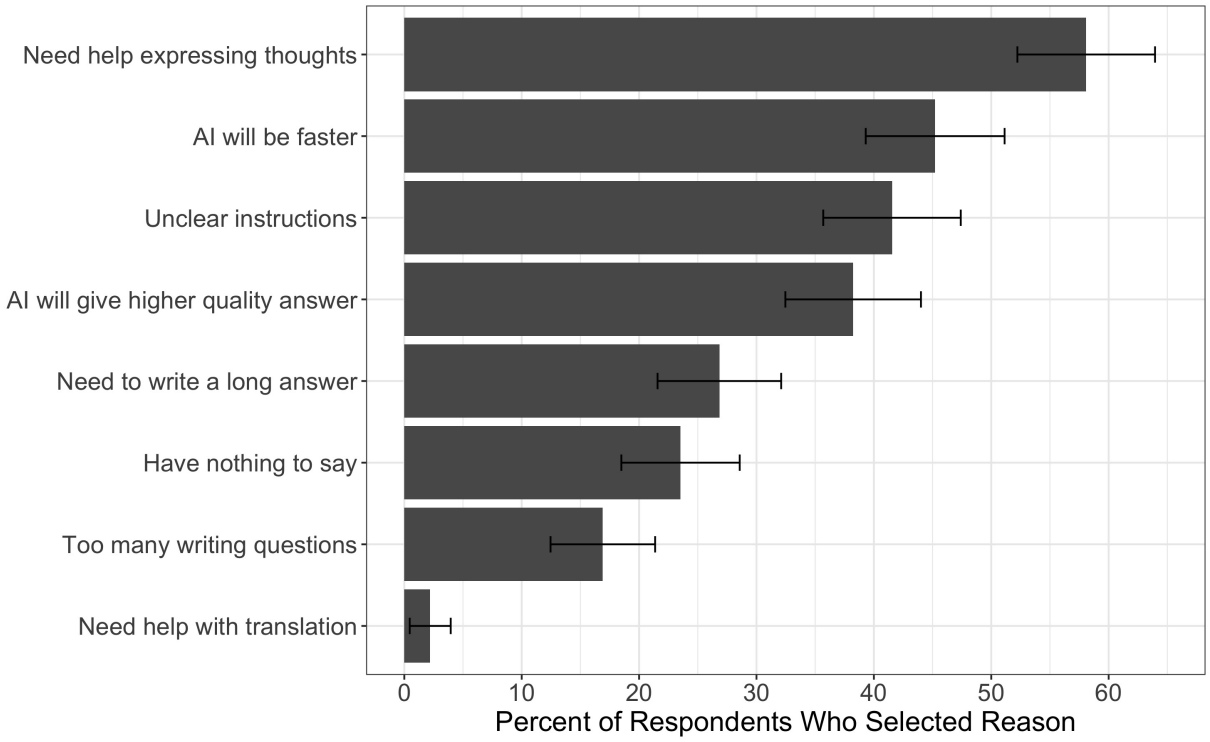
We asked respondents who said that they "sometimes", "most of the time", or "always" use LLMs to indicate all the situations that might lead them to turn to AI for help. The multiple-choice answer options included reasons related to perceptions of AI capabilities (e.g., AI would be faster, AI would give higher quality answers), needs related to the act of writing itself (e.g., expressing thoughts, elaborating longer answers, coming up with ideas for what to say, translation), and survey design shortcomings (e.g., unclear instructions, fatigue from answering too many writing questions). Conversely, respondents who reported "never" using LLMs were asked, "Why have you *not* used an AI assistant or chatbot to help you with survey questions that involve writing?" This question was originally posed as an open-ended question. After reading a randomly drawn sample of 100 responses per question, we inductively developed a coding scheme to manually categorize the respondents' answers. When a respondent's answer contained multiple reasons, the authors and research assistants labeled it with all the codes that applied.

Figure 3 displays the prevalence of self-reported reasons for using AI assistants, among respondents who use them at least sometimes. The most common reason – selected by 58% of AI users – is needing help expressing one's thoughts. Notably, while most respondents note that they turn to AI for help with articulating their opinions, they are much less likely to say that they turn to AI because they have *no* opinion on the matter or because they need to write a longer answer. In other words, respondents are more likely to say that they are using LLMs for help with verbalizing their ideas rather than using LLMs to generate new ones.

Some respondents also turn to AI when they think that generative AI might have stronger writing capabilities than they do personally. About 45% of AI users report using LLMs when they think that AI will be faster at writing an answer than they will be, and 38% say that they use AI when they think LLMs will generate a higher-quality answer than they would on their own. Finally, we find that some participants are turning to AI to compensate for shortcomings or fatigue caused by sub-optimal survey designs. Around 42% of AI users say that they use AI assistants or chatbots

when they encounter "unclear instructions", and 17% say they use AI writing assistance when they think the study includes too many questions that require writing.

Figure 3: **Reasons for Using AI** among respondents who reported using AI "sometimes," "most of the time," or "always" to complete open-ended writing questions. Respondents were asked to select all options that applied.



By contrast, Table 4 shows common reasons given by respondents who never use AI for avoiding it, along with descriptions, examples, and the percentage of respondents citing each reason (95% confidence intervals shown in parentheses). Most respondents' answers could be mapped onto a single reason category, but about a quarter of responses expressed more than one type of reason.

Table 4: Reasons Against AI Use

Reason	Description	Example	Prevalence (95% CI)
Inauthentic/ Invalid	Using AI would lead to inauthentic, invalid, or fake answers that do not reflect what respondent really thinks or believes	"It is not authentically my opinions or thoughts. Using an AI may change the tone and language I might normally use"	35.4% (31.3% -39.5%)
Unethical/ Immoral	Using AI would be unethical, immoral, or dishonorable in some way (i.e., cheating)	"I would still consider it a form of cheating and, therefore, believe it is unethical to use something that substitutes your own mind"	24.1% (20.5% -27.8%)
Unnecessary	Respondent does not see a need for AI	"Because I dont need to"	23.4% (19.7% -27%)
Against Rules	Using AI would violate rules of study participation or lead to platform sanctions	"I believe that it is against the rules to do so in most studies"	19.9% (16.5% -23.3%)
Preference	Respondent prefers to write their own answers instead of using AI	"I enjoy writing. If it's not a requirement, I prefer to write my own responses"	16.7% (13.5% -19.9%)
Unfamiliar	Respondent is unfamiliar with AI; doesn't know how to use it for completing surveys	"Honestly, I do not know how. It seems intimidating"	10.9% (8.2% -13.6%)
Ineffective	Respondent does not think AI is an effective tool	"Generative AI often spits out nonsense"	5% (3.1% -6.8%)

Note: Reason categories are not mutually exclusive

Overall, a mix of reasons emerged from the responses. The most common reason, expressed by over a third of the respondents who stated that they never use LLMs to answer open-ended questions, is concern about authenticity and validity. Responses in this genre often state that answers written with or by AI do not convey a person's genuine opinions, and evoke authenticity as a moral obligation that participants owe to researchers. Two other common reason categories also touch on moral boundaries: about 24% of responses note that using AI is unethical or immoral (i.e., it is "cheating") and about 20% of responses mention that AI usage violates rules and platform policies. Whereas concerns about authenticity draw attention to how LLMs may produce disingenuous answers, reasons mentioning ethics and possible rule violations surrounding AI usage highlight the perceived inherent "wrongness" of using these tools.

Beyond ethical concerns, other respondents avoided AI because they saw it as unnecessary or preferred their own writing process. About 23% of respondents said that they simply did not

need to use AI and about 17% said that they generally preferred writing by themselves. Many of the answers in these categories emphasized the respondent's competence, confidence, and/or enjoyment of writing. For these respondents, the ethics and implications of LLMs and their skill or familiarity with AI tools were less salient.

Finally, a small portion (about 11%) of respondents said that they don't use AI because they don't know how to, or because they are intimidated by the tools. An even smaller percentage (around 5%) reported refraining from using AI assistants because they don't think the technologies are effective, or because they demonstrated sophisticated understanding and distrust of LLMs.

4.3 Taxonomy of Prompting Strategies

We elicited sample prompts from respondents by showing them each of the three TESS survey questions of interest and asking, "Imagine you were participating in a study in Prolific and you encountered the following question. What would you say to an AI assistant or chatbot if you wanted to ask for help with writing a response to this question?" To reconstruct the question from the JSD study, we recreated the two different versions of the question that respondents were randomized into (i.e. "blacks" or "African Americans"). For questions from the Graham and Groenendyk studies, we first asked the relevant prerequisite questions and then tailored each survey question based on respondents' responses to these prerequisite questions (i.e. interested or uninterested in politics; respondents' own party vs other party). The three questions were shown in random order.

To maintain external validity, we only use prompts written by respondents who reported using AI assistants or chatbots at least sometimes. We developed a coding scheme of common prompt types by reading a random sample of 100 prompts per question. After identifying this taxonomy, we asked research assistants to classify the remaining cases with our code categories.

Table 5 shows the seven prompt strategies we identified, along with descriptions and examples, while Figure 4 shows their prevalence across each TESS survey question. In contrast to respondents' self-reported reasons for using AI for answering open-ended questions, which focused mostly on articulating and expressing one's own thoughts, the actual prompts rarely included infor-

mation about the respondents' own opinions or ideas. Instead, the most common prompt strategies were either idea generation (for the Groenendyk question) or information seeking (for the JSD and Graham questions). Less than 9% of AI users in our sample wrote prompts that included any perspective-taking information.

Table 5: Taxonomy of Prompting Strategies

Strategy	**Description**	**Example**
Idea Generation	Prompts asking for a list of items, topics, or ideas related to the subject. Generally open-ended and does not state the respondent's own position	"What are some reasons to be interested in politics?"
Information Seeking	Prompts asking for background information or answering a question as if it were fact-based	"How well do African Americans do in today's society?"
Writing Assistance	Prompts explicitly asking for help in formulating a response	"If I'm a Republican or Democrat what makes me different from the other? Please write a few sentences"
AI Opinion Solicitation	Prompts asking the AI what position it would take on a question, without asking it to take a particular perspective	"What is your stance on politics?"
Not Applicable	Respondent said they would answer that question themselves, or wrote something irrelevant	"I will write these myself, this is a personal question"
Direct Copy	Respondent directly copied and pasted the original survey question (or a substantial portion of it) as the prompt	"What makes you similar to Republicans? What makes you different from Democrats?"
Perspective Taking	Prompts asking AI to adopt a specific perspective	"act like a 45 year old basic white indifferent human, what do you think of african americans in american society today"

Note: Strategies are not mutually exclusive

Figure 4: **How often respondents incorporated different prompting strategies across TESS questions.**

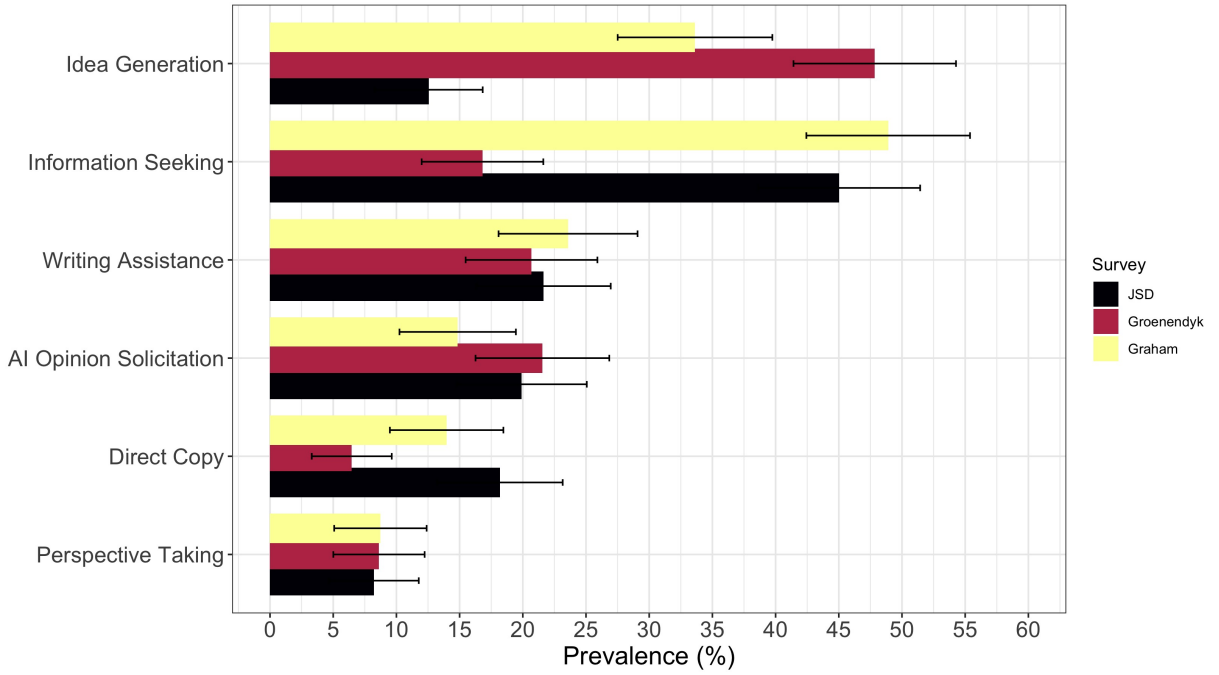


Figure 4 also shows that participants' prompt strategies vary substantially across survey questions. For example, in the two questions about intergroup attitudes (JSD and Graham), information seeking prompts were the most common. These types of prompts asked LLMs to bring in more contextual information about the group, or tried to reframe the question as a fact-based one, rather than focusing on one's own attitudes. On the other hand, for the more abstract question on interest in politics (Groenendyk), idea generation was a more popular strategy. Respondents were more likely to directly copy the JSD and Graham questions on intergroup attitudes than they were to copy the Groenendyk question on interest in politics, perhaps because the JSD and Graham questions are more sensitive. Although perspective-taking strategies were rare, their prevalence did not vary significantly across the three questions.

5 Analysis Part II: Comparison of Human vs LLM-Generated Survey Responses

In Analysis Part II, we compare human and LLM-generated survey responses to the three TESS questions along a number of dimensions, detailed in each subsection below.

LLM Survey Response Simulation Strategy: We simulate LLM responses using the prompts that respondents in our Prolific survey wrote for the three TESS questions.[2] We restrict to prompts by Prolific respondents who reported using AI assistants or chatbots at least some of the time for greater external validity. We manually reviewed all submitted prompts to screen out responses that were not valid prompts, such as responses that described in general terms how the respondent might prompt an LLM or that directly answered the underlying TESS survey question.

We simulated open-text responses using five LLMs: GPT-4o mini, GPT-4o, Gemini 1.5, Claude 3.5 Sonnet, and Claude 3 Opus. We focus on these models because they were among the most popular LLMs among respondents in our Prolific survey and because they provided API access to researchers. Our aim in the simulations was to approximate the LLM responses study participants might receive when using the public, free versions of ChatGPT, Gemini, and Claude. Accordingly, we present results based on the default models and model settings offered in free chat interfaces at the time we fielded the Prolific survey – GPT-4o mini (temperature = 0.7), Gemini 1.5 (temperature = 1), and Claude 3.5 Sonnet (temperature = 1) – in this paper's main analyses. To illustrate the robustness of our findings, we also present parallel results for GPT-4o mini with other temperature settings (0.0 and 1.0), GPT-4o, and Claude 3 Opus in the Online Supplement.

Not all LLM responses to Prolific respondent-written prompts were valid open-ended text responses. We filtered out any responses in which the LLM refused to answer the question, following safety protocols that the companies that developed the LLMs put into place. We also removed LLM

[2] In preliminary work, we also tested using the original TESS survey questions as prompts. The LLMs often refused to provide responses when prompted in this manner (e.g. from Claude: "I don't actually have any personal interest in politics. I'm an AI assistant created by Anthropic to be helpful, harmless, and honest. I don't have personal opinions or interests on topics like politics.")

responses that did not address the underlying question, such as responses that provided generic advice for giving a thoughtful response without providing any substantive direction. The rate of refusals and invalid outputs varied across models: 16% for Claude 3.5 Sonnet, 14% for Gemini 1.5, and just 1% for GPT-4o mini.

We further post-processed responses to remove additional text surrounding an open-ended text response, such as when an LLM opens with, "Certainly! I would be happy to help you with that." When initial LLM responses were not formatted as responses to a survey, we added a follow-up prompt that asked the model to revise the original responses to take the form of an answer to an open-ended survey question.[3]

Matching LLM-Generated Responses to Human-Generated Responses: To assemble a comparison set of human-generated responses, we match each Prolific respondent-written prompt with a corresponding human-generated response from TESS replication packages. We exactly match each Prolific respondent with a corresponding TESS respondent along a number of background characteristics. As all the questions we examine are broadly on political attitudes and beliefs, we chose to match respondents on several sociodemographic factors predictive of individuals' political views in the U.S. (Manza and Crowley, 2018; Jardina and Piston, 2019; Iyengar et al., 2019): race/ethnicity, education (BA or higher, less than a BA), partisanship (Democrat, Republican, or Independent), and age (45 and older or not). Since the TESS questions had randomized components (question wording in JSD) and/or components conditioned by earlier questions (interest in politics in Groenendyk; own and other political party in Graham), we further restricted candidate matches to cases where the Prolific respondent drafting the prompt saw the exact same version of the question as did the TESS respondent. Where there were multiple matches available, we randomly sampled one.

This procedure matches every set of LLM responses generated by a single Prolific respondent-drafted prompt with a single human-generated answer from the original TESS survey data. As we required exact matches, some Prolific observations had no matches in the TESS data and we

[3]The LLMs sometimes provided initial responses that were long, bullet-point lists, which we believed survey respondents would be unlikely to directly use as a response.

pruned those observations from the sample (there were no matches for 8 observations in JSD, 33 in Graham, and 42 in Groenendyk).[4] At the end, we had matched sets of $n = 223$ for JSD, $n = 195$ for Graham, and $n = 187$ for Groenendyk. All analyses that follow are based on these matched sets.

5.1 Response length and vocabulary size

We start by comparing the length of responses generated by humans and LLMs, as well as the size of their vocabulary (the total number of unique words that responses contain). Table 6 summarizes the average lengths and vocabulary sizes of text responses from our primary models (for Claude Opus and GPT-4o, see Table S1 in the Online Supplement) and from human respondents. LLM responses generally have larger vocabularies and tend to be longer. They also show greater variation in length than do human responses.

At first glance, this might suggest that LLM responses capture a wider variety of concepts and ideas than human responses do. Later analyses, however, more directly compare the amount of variation in the content of responses drafted by humans and LLMs, contradicting this initial interpretation.

Model	**Groenendyk**			**Graham**			**JSD**		
	MW	SD	TV	MW	SD	TV	MW	SD	TV
Claude 3.5 Sonnet	103.0	43.4	1859	101.4	35.1	1061	99.0	41.2	1522
Gemini 1.5	123.5	69.6	2131	161.1	95.7	1487	175.8	106.6	2358
GPT-4o mini	195.4	75.7	2690	215.8	101.1	1802	214.5	78.2	2527
Human	9.7	9.0	462	31.7	25.2	1146	16.1	25.4	915

Table 6: Comparison of word count and vocabulary size between human and LLM-generated responses. MW: mean # of words; SD: standard deviation # words; TV: total vocabulary.

[4] All 33 pruned Graham observations were of Independents in our Prolific sample, who were excluded from the original TESS study. The higher number of pruned observations for Groenendyk reflects the fact that the version of the question we fielded had a lower number of observations in the original TESS study, so there was a smaller pool of candidates for matching.

5.2 Most frequent and distinctive words

Beyond counting the number of words, we also examine which words appear more often in responses from humans and LLMs. Word frequency tends to follow a highly skewed distribution, where a small number of words appear much more often than other words. In English, this is especially true for function words like "the" and "of." We account for this in two ways. First, we remove common function words and other similar types of words using a standard list of stop words. Second, we use term frequency-inverse document frequency (tf-idf) to identify words that are distinctive to a particular set of responses. This method adjusts for how widely a word appears across all responses, assigning higher weights to words that are frequent in a single response but rare overall, and lower weights to words that are common across many responses. This approach allows us to compare distinctive vocabulary usage between human and LLM responses, highlighting the words that characterize each source's unique linguistic patterns.

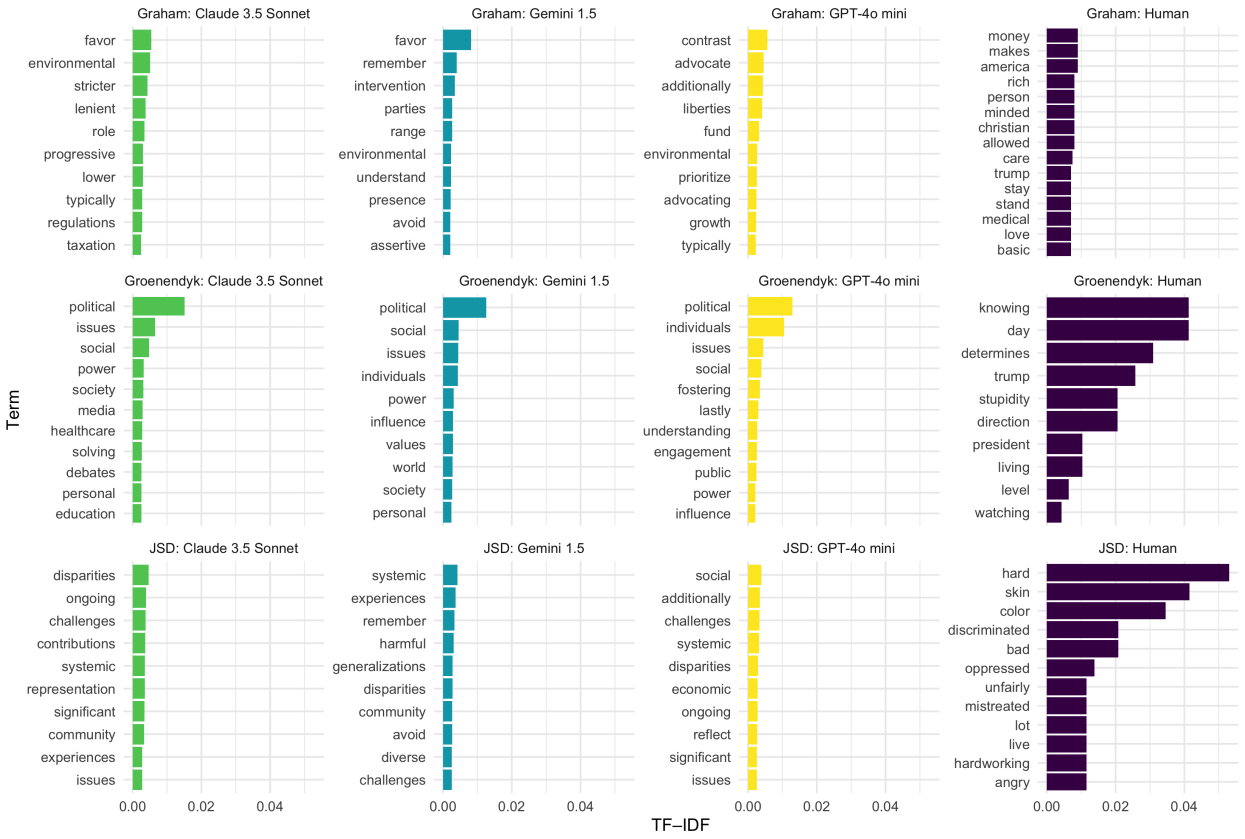


Figure 5: Highest TF-IDF terms in responses by text source. These words all appeared more than three times to account for potential outliers.

Figure 5 shows distinct patterns in language use across human respondents and LLM models. Human responses tend to contain more emotionally charged and concrete language. All three language models, meanwhile, consistently use more neutral, abstract language across all three questions. When discussing perceptions of Democrats and Republicans (Graham), LLMs focus on broader concepts like "environmental" issues and "liberties." For the question on political interest (Groenendyk), LLM responses center on terms like "influence," "issues," and "power," indicating a more analytical approach.

The contrast is perhaps clearest in responses to the JSD question on how the respondent perceives Black people in the US. Human respondents used adjectives like "oppressed", "discriminated," and "hard," suggesting a focus on concrete experiences and discrimination. LLM responses, meanwhile, answered with more abstract nouns and adjectives like "disparities", "sys-

temic," and "challenges."

Overall, this difference in language use suggests that LLM models may approach sensitive topics with more detachment, while humans tend to express stronger opinions using more concrete, everyday language. This general pattern also holds for other models, as shown in Figure S1 in the Online Supplement.

5.3 Response sentiment

We compare the overall sentiment expressed in the language of human- and LLM-generated responses using the Valence Aware Dictionary and sEntiment Reasoner (VADER) lexicon, a popular dictionary-based method that assigns words as being either positive, negative, or neutral in sentiment (Hutto and Gilbert, 2014). After each word in a response is tallied as either being positive, negative, or neutral, VADER generates an overall composite score for each response. For simplicity, we present the degree of positive and negative sentiment expressed in responses.

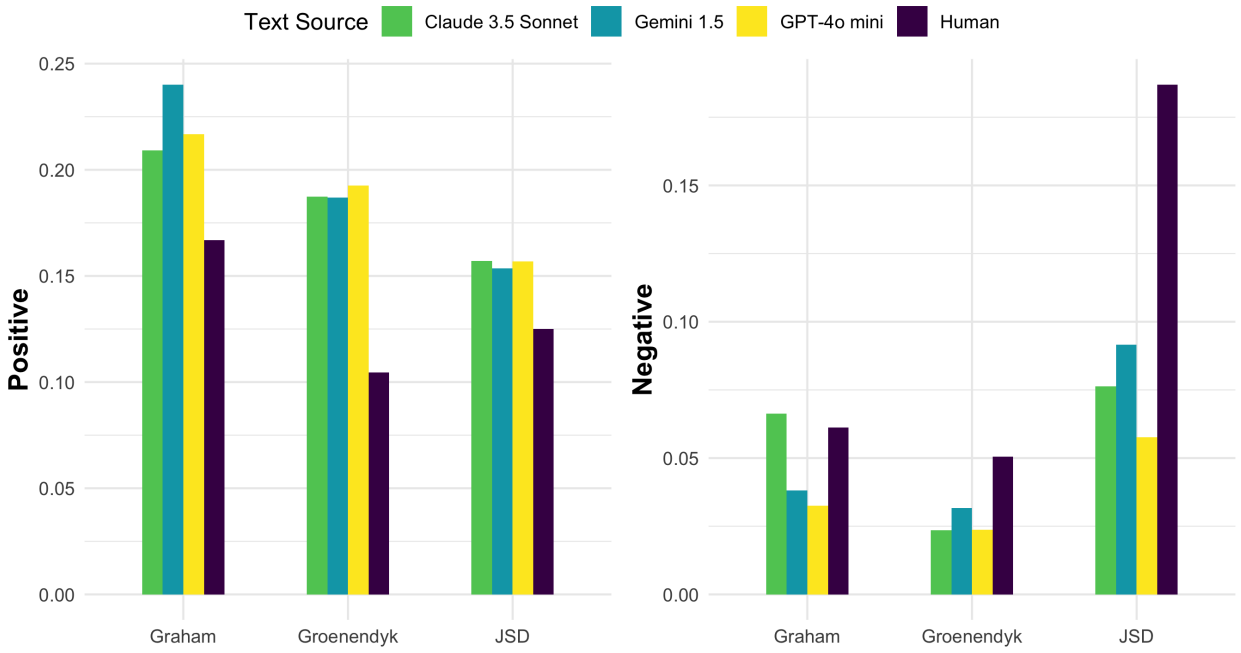


Figure 6: Average positive and negative sentiment scores for each set of responses as measured using VADER.

Overall, LLMs used more positive language (lefthand side of Figure 6) and humans used more negative language (righthand side of Figure 6). The contrast was particularly pronounced in responses to the Groenendyk question about political interest, where LLMs used nearly twice as many positive words as humans (around 0.18 - 0.19 for the three LLMs vs. 0.10 for humans). The sentiment divergence was also striking in responses to the JSD question on perceptions of Black Americans, where humans used more than twice as many negative words per response as LLMs.

We also found similar patterns among the models presented in the Online Supplement (see Figure S2).

5.4 Use of dehumanizing language

Two of the questions we examine focus on perceptions of social groups: Black people in the JSD survey and Republicans/Democrats in the Graham study. In this section, we dig more into the nature of the language that responses employ to describe other people. We focus, in particular, on the use of dehumanizing language, which has been a subject of broad interest to scholars given the wide array of social and political concerns that the dehumanization of others raises (Hagan and Rymond-Richmond, 2008; Haslam and Loughnan, 2014; van Loon et al., 2024).

Following prior work on measuring dehumanizing language (Mendelsohn et al., 2020), we assess the degree to which the language used to describe social groups reflects moral disgust and vermin-related metaphors. To do so, we measure semantic similarity between the responses generated from each source (human, GPT, Claude) and the two concepts (moral disgust and vermin) using pre-trained GloVe (Global Vectors for Word Representation) word embeddings (Pennington et al., 2014). Word embeddings represent words as numeric vectors that capture semantic relationships, allowing researchers to quantify the similarity between different words or concepts based on their usage patterns in existing large text corpora such as Wikipedia (for reviews, see Kozlowski et al., 2019; Stoltz and Taylor, 2021; Arseniev-Koehler, 2022).

We constructed word embedding vectors to represent each concept, following the approach in Mendelsohn et al. (2020). For vermin, we averaged the GloVe word embedding vectors for a list

of vermin-related words (e.g. rodent(s), rat(s), cockroaches). For moral disgust, we averaged the word embedding vectors for the words in the Moral Foundations Dictionary reflecting the concept (Frimer et al., 2019). The Moral Foundations Dictionary provides a lexicon with which to analyze text for content related to various moral dimensions, including moral disgust, as elaborated in Haidt and Graham (2007).

We then represented each human or LLM-generated response in terms of its average word embedding. First, we removed stop words, tokenized the text, and calculated term frequencies for each response. We then loaded the GloVe embeddings (100-dimensional vectors) and computed the weighted average of its constituent word vectors, with weights proportional to term frequencies. This process resulted in a single 100-dimensional vector representing each text's semantic content. To quantify the similarity between each response and the concepts (vermin, moral disgust), we calculated the cosine similarity between each text's average embedding and the vector for each concept. Cosine similarity in this context provides a measure of semantic closeness, with values ranging from -1 (perfectly dissimilar) to 1 (perfectly similar).

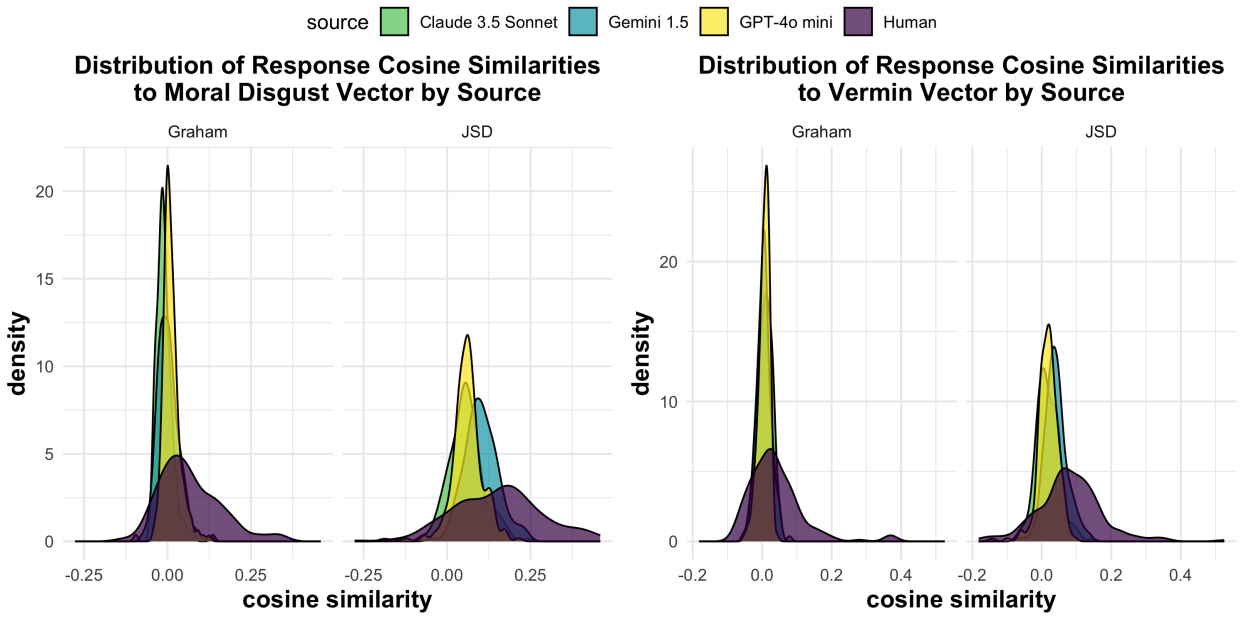


Figure 7: Distribution of response cosine similarities to pre-trained GloVe word embedding vectors for words related to vermin and moral disgust

Figure 7 illustrates the distribution of response cosine similarities to the moral disgust vector

(left panel) and the vermin vector (right panel) across the three sources of text.

The results reveal a consistent pattern: AI-generated responses from GPT-4o mini, Gemini 1.5, and Claude 3.5 Sonnet cluster more tightly around a lower cosine similarity to the vermin and moral disgust vectors than do human responses. Whereas human responses reflect a wide range of perspectives and language use, AI responses tend to be more uniform. Particularly noteworthy is that a substantial proportion of human responses included more dehumanizing language in their written discussions of Black Americans and Democrats/Republicans than did the LLMs. Consistent with the results of the preceding sentiment analysis, these findings indicate that LLM-generated answers truncate the range of views of social groups that exist among people, particularly those of a more denigrating or negative nature.

The Online Supplement shows that this general pattern of results holds for Claude 3 Opus and GPT-4o, as well as for other moral dimensions in the Moral Foundations Dictionary such as fairness/cheating and authority/subversion.

5.5 Response variation/homogeneity

In this section, we more directly examine whether LLM-generated responses are more homogeneous than human-generated ones. We represent each response as a vector of tf-idf scores and then measure how similar these vectors are to each other using cosine similarity. We calculate the average pairwise cosine similarity between responses from the same source (i.e., humans to humans, GPT to GPT) as a proxy for text homogeneity. If the average similarity for responses from a given source is high (closer to 1), then responses from that source tend to be more homogeneous in their word choices.

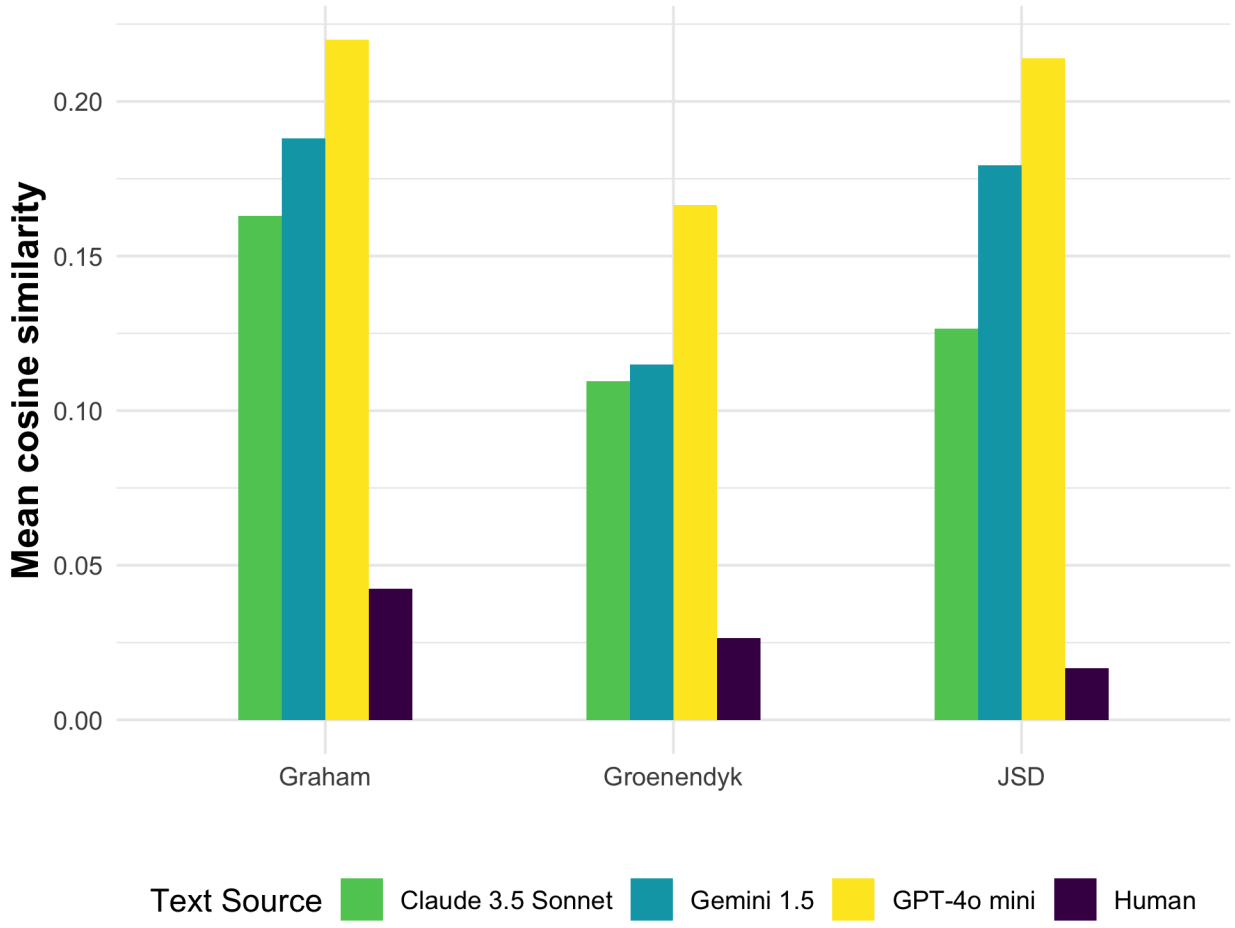


Figure 8: Average within-group pairwise cosine similarity of tf-idf vectors for responses generated by humans and LLMs.

As Figure 8 shows, for all three questions, the average pairwise cosine similarity between responses generated by the LLMs was higher than for responses generated by humans. In other words, LLM-generated responses tend to be more similar to each other than human-generated responses are to other human-generated responses. Responses from GPT-4o mini had the highest mean cosine similarities.

These results held for other models and temperatures that we present in Figure S6 in the Online Supplement. Notably, even though we used a broad range of prompts sourced from online survey respondents to reflect variation in peoples' prompting approaches, LLM responses were still more homogeneous. This was also the case despite the LLMs generating much longer responses with larger vocabularies than did humans. The fact that this problem is more acute for GPT, by far the

most popular tool among our respondents, further underscores this challenge.

6 Conclusion

Summary of findings: In this paper, we examine how online survey participants use AI tools to complete open-ended survey questions and compare how LLM-generated responses differ from answers written by humans. Drawing on an original survey of online survey participants, we estimate that about 34% of online study participants use LLMs to answer open-ended questions at least some of the time and find that OpenAI's ChatGPT is the model of choice for over 90% of all LLM users. We also find that LLM usage varies by race, gender, education, political party affiliation, and survey-taking experience, which could compound data quality issues caused by systematic sampling error.

Most LLM users reported using AI assistants or chatbots for help with expressing what they want to say, with few reporting that they use AI because they don't have any opinions of their own on the matter. However, in actual prompts submitted by users, the most common strategies involved asking AI for help with generating ideas or seeking additional information on the topic. Few prompts explicitly incorporated respondents' pre-existing views or social positions. This discrepancy raises questions for future research about how participants view ideas generated by AI during co-writing – do they view them as extensions of their own thoughts, or do they see LLM suggestions as "new" ideas? Finally, we find that participants who refrain from using AI tools often cite moral reasons (e.g. AI compromises authenticity, using AI is unethical and akin to "cheating", using AI violates rules and terms of services).

Using TESS survey data collected before the public release of ChatGPT and LLM-generated answers simulated using respondent-written prompts from our original survey, we find that LLM-generated responses tend to be longer yet exhibit greater similarity to each other than human-written responses. Compared to human respondents, LLMs also use more positive language and are less likely to employ dehumanizing language when discussing social groups. Overall, our re-

sults suggest that even when LLM prompts are customized to specific questions and reflect the varied prompting strategies of real users, LLM-generated text reflects a narrower range of perspectives and language use than human-written text.

Limitations: This paper has several limitations. One key limitation is that we do not examine actual AI-assisted survey answers. Instead, we compare human-generated responses to simulations of AI-generated answers. While we tried to increase external validity by generating LLM responses using prompts written by real study participants who use AI, participants may still further revise, selectively use, or reject AI's suggested answer. There is likely some slippage between the fully AI-generated responses we examine here and the AI-assisted responses survey-takers would actually provide. Thus, it may be helpful to interpret our results as reflecting the prospective answers that participants using AI are exposed to, which some participants may use directly and others may be steered by (Jakesch et al., 2023). Future research should more directly examine how study participants engage with AI-generated answers, following the example of other work that has studied how people interact with AI writing assistance in other types of writing tasks (Lee et al., 2022; Padmakumar and He, 2023; Sharma et al., 2023; Dhillon et al., 2024; Doshi and Hauser, 2024).

Another limitation of this work is that our comparison of LLM- and human-generated answers relies on matching respondents across samples fielded at different moments in time (2019-2021 for TESS; summer 2024 for our Prolific sample) along a set of important, but limited observable characteristics. Accordingly, our human-written answers may be imperfect counterfactuals for LLM-generated answers. For example, views of Black Americans as studied in JSD and Republicans/Democrats in Graham may have evolved over this time period. Nonetheless, such shifts alone are unlikely to account for many elements of our results, most notably the substantial divergence in the degree of variation between responses generated by LLMs and humans that we document consistently across a range of measurement approaches. Future research could further build on our results by implementing research designs that could provide tighter control, such as fielding a lab-based study in which participants are monitored and randomly assigned to either have access

to AI tools or not when answering survey questions.

A further limitation is that our data on AI use and respondent-generated prompts were based on self-reports. Since participants knew that they were in a study on AI, social desirability bias and demand effects could have influenced our data. Emerging research suggests that people hide their AI use to avoid being perceived negatively by others (Slack, 2024), suggesting that our self-reported usage measures likely underestimate true AI adoption. Nonetheless, future research could couple self-reports with computational approaches looking for linguistic markers of LLM writing (e.g. Liang et al., 2024; Veselovsky et al., 2023b), though such strategies may prove less reliable as models and user behavior continue to evolve. How self-reporting might bias respondent-reported prompts is more unclear. While a true behavioral measure would have been more externally valid, it would be far more difficult to obtain and our data on user prompting strategies still offer novel insights into user behavior. Future research could incorporate better behavioral data, such as digital archives of prompts or lab studies that track prompting behavior in a more fine-grained manner.

Our findings are also limited by the particularities of our survey sample. Our survey covers one commonly used platform – Prolific – but researchers also recruit online participants through other means like targeted advertisements, mailing lists, and other crowd-working platforms. Existing research suggests that AI usage among participants on Prolific is actually substantially lower than on MTurk (Christoforou et al., 2024), but additional comparative research is needed on AI usage across platforms to inform recruitment strategies for studies where AI poses a significant threat to validity.

Finally, our survey only captures a particular snapshot in time. As AI technologies continue to diffuse rapidly (Bick et al., 2024) and companies increasingly build AI into popular websites, digital applications, and operating systems, we expect that lowered barriers will lead more survey participants to use them for writing assistance. Prevalence among subgroups is also likely to shift. For example, we find that college-educated participants are more likely to use AI than those who do not have a four-year degree. However, this could be because college-educated knowledge workers are more exposed to LLMs through work and the education gap will disappear or even reverse

as more people engage with AI tools. Patterns of platform churn also play an important role in determining the future prevalence of AI-generated answers. In our survey, participants who joined Prolific in the last six months were more likely to use AI tools. If newer participants generally have a higher tendency to use LLM tools, then their entrance to the platform would continuously drive up average AI usage, especially if longer-term participants are more likely to exit than newer participants. Future research should examine the link between AI adoption and participant selection into survey platforms, and further explore how platform participants' LLM usage compares to that of the broader population.

Contributions: This article offers a portrait of when and why research participants employ AI when participating in online surveys and experiments. Our survey complements prior work inferring the incidence of LLM usage (Veselovsky et al., 2023a,b; Christoforou et al., 2024) by further contextualizing such practices in the perspectives and experiences of study participants themselves. Following other work aiming to draw lessons from the views of crowd-working platform participants (e.g. Kittur et al., 2013; Fowler et al., 2023; Reynolds and Kincaid, 2024), our study sheds light on factors that can produce misalignment between researchers' goals and research subjects' study participation strategies. By understanding participants' motivations and constraints, we identify opportunities for researchers, detailed in the following subsection, to improve both research data quality and the experiences of online research subjects.

AI use among crowdworkers, first identified by Veselovsky et al. (2023a,b), has become an increasingly salient concern among machine learning researchers and computational social scientists who rely on crowdworkers for data annotation. This study, alongside Christoforou et al. (2024), is among the first to directly elaborate on the implications of this development for social science surveys and experiments. Unlike data annotation tasks, in which participants provide ground truth or gold standard examples according to external guidelines – serving more in the role of research assistants at scale – surveys and experiments are fundamentally aimed at gathering valid information on human subjects themselves. This article translates challenges that AI usage raises for survey and experimental researchers, situating this emerging trend in prior literatures on ensuring

data quality in online samples.

An important line of research has explored the positions and perspectives that LLMs tend to output when prompted in different ways. To understand how these LLM tendencies might shape survey responses, we gathered data on the types of prompts participants report using, showing that participants rarely prompt AI in the detailed, tailored manner that researchers do to replicate human responses. Instead, they more often draft more generic prompts, which our simulations show tend to generate more homogeneous, positive answers to sensitive questions about inter-group perceptions than human respondents normally write. In so doing, this study adds to scholarship that bridges research on LLM biases and capabilities with their downstream societal consequences (e.g. Jakesch et al., 2023; Potter et al., 2024).

Implications and practical recommendations for online research Our findings on why respondents use or avoid AI suggest several strategies to discourage or strategically manage LLM use in online studies. Many participants cited ethical concerns, a desire for authenticity, and a wish to follow the rules or researchers' preferences as reasons for not using AI. This suggests that explicitly asking respondents to avoid AI, making normative appeals about the importance of authentic responses, and fostering participant buy-in could help reduce AI use. Indeed, Veselovsky et al. (2023a) found that directly asking participants to refrain from using AI reduced its use, especially when combined with measures that added friction, such as preventing copy-pasting. Researchers may also consider adding study elements aimed at detecting LLM influence that could be used to filter responses, such as instructions that exploit known LLM tendencies or weaknesses. We note, however, that researchers should carefully weigh the trade-offs of such an approach as it may raise similar issues as traditional attention checks (Berinsky et al., 2014; Muszyński, 2023), such as falsely flagging genuine human-written responses as AI-produced, confusing respondents, and introducing selection bias.

Our results additionally indicate that overly demanding writing tasks or unclear instructions can prompt participants to turn to AI. To mitigate this, researchers should continue to follow long-standing best practices, such as thoroughly pretesting and streamlining their studies to reduce re-

spondent burden. As respondents may find the act of writing particularly burdensome (Fowler et al., 2023), another possibility could be to leverage speech-to-text models to allow respondents to answer questions orally. Researchers should exercise caution, however, as such models may fabricate information or introduce sociodemographically disparate transcription errors (Koenecke et al., 2020, 2024).

We further emphasize that not all AI use is necessarily detrimental: the most commonly cited reason for AI use–help with expressing oneself– suggests that AI could be beneficial in some contexts. Researchers may consider establishing guidelines for LLM use where it aligns with their research goals. They may, for instance, encourage respondents to first outline their thoughts in their own words prior to turning to AI to refine their answers, an approach that could help better anchor responses in participants' authentic perspectives. Researchers could also track how AI informs responses by providing space for participants to enter their initial answers and AI-assisted answers. They could even build AI writing assistance directly into survey and experimental instruments, enabling a more detailed look at how AI shapes responses to their questions. Future studies should assess which models of AI writing assistance best further researchers' goals (Lee et al., 2024). More broadly, future research should evaluate the effectiveness of the various interventions identified here and clarify the conditions under which they are most productive.

In drawing implications from our findings, we additionally stress that only 9% of respondents indicated that they use LLMs "most of the time" or "always" when answering open-ended questions; the remaining 25% only indicated that they use LLMs "sometimes." While AI use may increase over time, at the time of writing, we caution against interpreting our findings as undermining all open-ended text data collected in recent studies. Our results suggest that both the proportion of and degree to which responses are AI-influenced will vary substantially across studies. Our survey results suggest that AI use may be more extensive in writing-intensive questions on topics that are challenging to discuss or where respondents have limited knowledge or weaker opinions.

To gain a better sense of how AI usage could influence a study, we recommend that researchers proactively test their open-ended questions by simulating responses using commonly employed AI

models, following the approach we took in this paper. Researchers could examine how often AI tools refuse to answer their questions and explore what tendencies LLM-generated answers exhibit in their research setting. With this knowledge, researchers can refine their question phrasing, anticipate the direction of potential biases introduced by AI assistance, or fine-tune models to detect AI-generated or AI-influenced responses to their specific questions. Additionally, researchers can use this information to conduct sensitivity analyses that evaluate how much AI-induced bias would be required to overturn their findings. We suggest that implementing such measures could become standard practice amidst the growing take-up of AI assistance, helping producers and evaluators of research alike more precisely assess whether and how AI use affects the validity of specific research findings.

Efforts to align AI systems with human values and train them to avoid producing biased or harmful content are crucial for managing AI risks and ensuring societal benefit (Kirk et al., 2021; Ouyang et al., 2022; Kirk et al., 2024). However, these safeguards can have complex consequences when research participants use AI tools to formulate responses, especially to sensitive questions. If LLMs are trained to avoid generating content that could be seen as offensive, they may steer participants' writing toward more sanitized and less authentic expressions of their opinions and experiences (Jakesch et al., 2023). This issue may be particularly acute for questions related to inter-group attitudes and perceptions, which are of significant interest to social scientists. When LLMs generate responses to such questions, they often lean toward generating more positive and flattering depictions of social groups, potentially muting ambivalent or negative perspectives. LLM answers indeed stood in sharp contrast to human responses, which featured a wide range of opinions, including highly derogatory ones. Researchers should consider these AI-induced influences when designing studies and interpreting their results.

Our results on the increasing use of AI among online study participants and how it may influence research findings further point to the growing need for research designs to incorporate additional validation steps or triangulation between different research methods and data sources. Indeed, traditional qualitative methods like interviews and ethnography may be more critical than

ever for generating accurate knowledge and understanding of patterns in human beliefs, attitudes, and experiences (Grigoropoulou and Small, 2022; Small, 2024).

Implications for work on AI evaluation and societal implications of AI: Our finding that a substantial share of online study participants use generative AI tools to respond to open-ended survey questions introduces an additional concern for scholars evaluating how well AI is able to predict or mimic human perspectives: the risk of contamination. If AI-influenced responses are used to evaluate AI's ability to replicate human thoughts and judgments, it creates a feedback loop that can undermine the validity of assessments (Veselovsky et al., 2023b). We might mistakenly conclude that AI systems are better at reproducing human views than they really are. This issue also extends to computational social science research that relies on human-labeled data for supervised learning tasks. If a significant portion of the data used to train these models comes from crowdworking platform participants who used AI to generate or assist with their responses, it becomes increasingly difficult to establish a true human baseline or "ground truth" against which to evaluate model performance.

Furthermore, over the long term, there is a risk of a self-reinforcing cycle, where survey data and results containing AI-influenced content are used to train models, which in turn power AI tools that influence the responses of future study participants (Shumailov et al., 2024). Over time, this could lead to a further homogenization of perspectives and a narrowing of the range of ideas reflected in research data. Even beyond the problems this would pose for social science, these findings also point to everyday implications for the ways that LLMs may standardize our understandings of the social world (Evans and Foster, 2024; Foster, 2023). To mitigate these risks, researchers, practitioners, and crowdsourcing platforms should prioritize the development of methods and standards for detecting, monitoring, and managing AI-influenced writing products. Cross-disciplinary collaboration will be critical for understanding, addressing, and managing these emerging challenges.

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